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PI Name:	Boyle, Richard Ph.D.		
Project Title:	Structure & Function of Neurovestibular Adaptation & Re-adaptation in the Utricle Following Extended Periods of Hypergravity Exposure		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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PI Email:	richard.boyle@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1099
Organization Name:	NASA Ames Research Center		
PI Address 1:	Building N239, Rm 205		
PI Address 2:	RESEARCH AST, BIOLOGICAL STUDIES		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
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Key Personnel Changes/Previous PI:	No key personnel changes during this reporting period		
COI Name (Institution):	Kondrachuk, Alexander Ph.D. (Dept. Theoretical Physics, Institute of Physics, Natl. Acad. Sci., Kiev, Ukraine)		
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Task Description:

The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite sense.
In the utricular otolith organ it was found that the afferent response sensitivity to head acceleration (13, fish) as well as the number of synaptic bodies in hair cells (56, rats) increase as the result of exposure to microgravity (μG). A preliminary result showed that exposure to hyper-G by centrifugation decreases the number of synaptic bodies (56, rats). The period of afferent re-adaptation to normal gravity in fish (13) was found to be close to the period of behavioral adaptation in humans (55). Re-adaptation is defined here as the process of adaptation to normal gravity. These results suggest the possibility that the process of adaptation to μG might be mimicked by the process of re-adaptation from a hyper-G level. However, the experiments that would combine the results of μG and hyper-G experiments for the same species are absent. Further, the correlations of the synaptic re-arrangement with afferent sensitivity following exposure to altered gravity for the same species remain undefined. The purpose of the present proposal is to fill these gaps.

Our model is the adult toadfish, Opsanus tau. Fish provide an excellent model system for gravitational research (7). We have everything in place to successfully complete the proposed set of experiments and modeling. Of prime importance is the fact that in the same species a comprehensive evaluation of the neural response to μG (13) and hyper-G (present proposal) exposure will be made. There are four specific aims:

Aim 1. Response characteristics of utricular afferents to Centrifugation. Fish will be placed in groups and exposed to different magnitudes (1.5 and 3 g) and duration (1-16 days) of centrifugation. On Center Controls and ground controls are included. Using electrophysiological techniques, individual utricular afferents will be physiologically characterized based on gravitoinertial acceleration testing. Afferent response properties to efferent vestibular activation will be specified. The duration of afferent recording session is tempered by the need to freeze the tissue in time to permit the measurements of structure-function correlations. To match the conditions under which post-flight experiments were performed (13) changes in afferent sensitivity caused by hyper-G will be studied from 1-24 hrs after centrifugation, and re-adaptation after 24-96 hrs in 1G .

Aim 2. Morphology of hair cell-afferent synaptic complexes. Combining the techniques of labeling of individual afferents with light and transmission electron microscopy a complete characterization of the afferent is possible: its physiology, its dendritic morphology in the macula, and the extent and quantification of synapses formed from the innervated hair cells. Visualization of complex 3-D microcircuitry will also reveal precise connectivity within the macula that is essential for understanding neurovestibular adaptation. Synaptic body counts will be performed to address the fundamental question of the correlation of afferent response sensitivity with hair cell synapse number.

Aim 3. Merging the structure and function of otolith afferents and their relationships to hyper-G. The specific times and thresholds of the afferent response adaptation and re-adaptation and the structural changes associated with the hair cell-afferent synapse will be determined by varying the magnitude and duration of centrifugation.

Aim 4. Modeling. Based on known methodologies and the present experimental results, research will be directed towards the establishment of phenomenological models describing the adaptive changes in utricular afferents in altered gravity conditions. We will examine the relationship between the afferent response and the number of synapses in normal and in hyper-G conditions. A model of toadfish otolith mechanics in these conditions will be also developed.

The four specific aims are directed to achieve the main goals: 1) to test the correlations of change in the gravitational level and the change in afferent response characteristics caused by head acceleration and the potential underlying changes in the morphology of the associated hair cell-afferent synaptic complexes, 2) to examine the role of efferent vestibular system on these correlations, 3) to test (propose) the possible mechanisms responsible for these correlations; and 4) to test if the transition from hyper-G to normal gravity can be used to mimic the effects of neural adaptation to μG.

We expect that convergence of findings in μG and hyper-G will provide a model of neurovestibular plasticity, and will lead to the development of novel test protocols for both ground- and space-based research. As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness or even modify synaptic sensitivity in preparation of entry into a higher gravity level.

The overarching hypotheses to be evaluated are: One, the gravitoinertial sensitivity of otolith nerve afferents is altered as a consequence of the animal's exposure to hyper-G conditions as well as μG. It is reasoned that the transition from 1G to hyper-G will decrease otolith afferent sensitivity (output in impulses per sec/input in G), and the transition from hyper-G back to 1G will increase it (and back to normal levels). Thus, the transition from hyper-G to normal gravity may serve as an imitator of neurovestibular sequelae to μG. Two, hair cell-afferent synaptic complexes are modified by the altered gravity condition: the up- or down regulation of synaptic bodies is specific to the direction of transition from 1G. The change in number of synapses is directly correlated with afferent response sensitivity. Three, afferent adaptation mechanisms may involve either self-learning processes confined to the hair cell or responses of the hair cell-afferent system to feedback signal(s) that originate from a misbalance of information coming from otolith and extra-otolith contributor(s), or a combination of both. (These mechanisms will be considered by modeling based on the experimental findings.) Four, efferent vestibular actions are not uniform on hair cells and afferents, and it is proposed that the changes brought on in otolith afferents by the transitions from one gravity state to another will be most prevalent and coincident in afferents strongly affected by efferent activation. If this proves to be the case, then the next step is to directly query the efferent neurons themselves.

The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vertebrate vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae, and for the effective design and validation of countermeasures to benefit astronaut health and performance. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite

	sense. This study is a natural analog to our previous studies aboard STS-90 and 95, and will provide a comprehensive evaluation of the neural response to μG and hyper-G in the same species. We expect that convergence of findings in μG and hyper-G will provide a solid model of neurovestibular plasticity, and will lead to the development of novel test protocols for both ground- and space-based research. As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness, benefit sensorimotor coordination and performance, or even down regulate synaptic sensitivity in preparation of entry into a higher gravity level.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The functional importance of head stabilization is seen by our ability to change gaze orientation and maintain binocular fixation of objects of interest during not only standing but also during locomotor activities, and our ability to detect self-motion and adaptively react to applied perturbations of the head and body. In the space environment the vestibular system is profoundly affected by the loss of he gravity vector. However, conditions on Earth can seriously affect the individual's ability to functional normally. In addition aging places a serious challenge to the vestibular and neural centers responsible for posture and movement control. Disorders of the vestibular system affect head, gaze and posture stabilization, and can dramatically alter the patients willingness to induce self-motion and can lead to trepidation of normal activities. These studies will also provide important insights into vestibular function, and may help in the diagnosis and treatment of vestibular patients seen in the clinic.
Task Progress:	Task Report : “Structure & Function of Neurovestibular Adaptation & Re-adaptation in the Utricle Following Extended Periods of Hypergravity Exposure” Graduate Student Record: 1) Alireza Mofrad, June 2005 – present. Masters of Science, Department of Biology, San Jose State University. Medical Student Record: 1) Reza Ehsanian, B.S., M.S. 2004-present. Stanford Medical School, Palo Alto, CA. Postgraduate Record: 1) Yekaterina Popova, MD, PhD. January 2005 – present. Postdoctoral fellow of the Educational Associates program. Summer Interns 2007: 1) Sammy Park, University of Michigan, Aerospace Engineering 2) Paul Swift, University of Arizona, Chemical Engineering 3) Monique Chambers, University of La Verne, pre-Medicine 4) Emily Gabranski, University of New England, Biology Summer Interns 2008: 1) Michael Hrynyk, McMaster University (Hamilton, Ontario), Biology & Pharmacology 2) Long Nguyen, Stanford University, Human Physiology 3) Kaila O’Neill, Lyçée Français La Pérouse 4) Vikas Yendluri, Bellarmine College Preparatory 2007 VISITING FACULTY: 1) Alexander Kondrachuk, Ph.D., Senior Research Scientist, Group Leader, Dep. of Theoretical Physics, Institute of Physics, National Academy of Sciences of Ukraine, Kiev, Ukraine. Biophysical modeling of the utricular transduction mechanisms. 2) David Bruck, Ph.D., Professor, Dept. Biology, San Jose State University. Task Progress: We have made significant progress towards reaching our stated goals. Two separate and successful centrifugation trials were conducted to examine the response of utricular nerve afferents to 3g and 1.5g hyper-gravity stimulation and the time course of re-adaptation to Earth's 1g following centrifugation. The centrifugation stimulus and configuration have worked perfectly and without incident. Further, we have been able to successfully record the electrophysiological responses of more that 60-70 separate afferents in each experiment, twice what we had anticipated. This increase in sample has provided a very reliable measure to evaluate response between experiments. In addition, we designed a separate rate table to conduct the On-Center Control experiments. This design worked perfectly, allowing the subject to be rotated at 38 rpm in a control aquatic system with stabile water temperature of 12°C. As of October 2008, four (4) fish were studied following 3G exposure at 1, 2, 3, and 8 days; five (5) fish at 4 and 16 days; and three (3) fish at 5, 24, and 32 days (total 32 fish). The 24- and 32-Day groups were added to our protocol based on our results at 16-Day exposure to 3G. For the Re-adaptation experiments, three (3) fish were studied following 4-days of 3G exposure and re-adaptation periods of 1, 2, 4, 6 and 8 days (total 15 fish). In the next group of Re-adaptation experiments, three (3) fish were studied following 16-days of 3G exposure and re-adaptation periods of 2, 6 and 8 days, and four (4) at 4-days of re-adaptation to 1G (total 13 fish). We successfully completed four (4) fish exposed to 1.5G at 4-days and 16-days (total 8 fish). We successfully completed On-Center Control experiments of 38 rpm for 4- and 16-Day with 4 fish in each group. Results and conclusions based on the analyses conducted to date on the effect of altered gravity levels on utricular afferent response. Data were collected from 2 orbital shuttle missions: STS-90 and STS-95. Within the first day after landing, the magnitude of response to an applied acceleration was significantly greater than for controls. The number of synaptic bodies in type II hair cells in rodent is labile, increasing following exposure to microgravity, suggesting an increase in synaptic density was responsible for the upregulation of sensitivity. 3-g Centrifugation data. The data must be viewed still as preliminary, because not all data obtained have been analyzed. The mean maximum sensitivity (Smax) of control afferents to a standard translation is 2333 ± 1460 (SD) imp/s/g . The Smax for each group based on days of 3-g exposure is given. A significant elevation at 4-Day and a significant decrease at 16-Day ($p<0.0001$) of Smax were observed. This is an important finding: the response of the utricle to hypergravity is modifiable and bi-modal. With 4 days of recovery (Re-adaptation) following 3-g exposure for 4- and 16-Days the gain did not return (fall or rise) to control levels. Percent plot shows the wide separation in Smax between 4-Day (red circles) and control fish, and the response of ~80% of afferents in16-Day group (blue diamonds) diverged from control afferents. Length of time needed in the 1-g environment following 3-g exposure of 4 and 16 days for afferents to return to baseline levels. Only partial analysis is complete and recovery times of 1, 2, and 6 days have not yet been analyzed. The general conclusion is: the hypersensitivity within the first days of 3-g exposure and the hyposensitivity observed after longer durations of 3-g exposure require at least 4 or more days of return to 1-g to complete the readaptation process. It should

	be stated again that horizontal canal afferents responded in a normal fashion within each of the experimental groups following 3-g exposure with or without 1-g recovery days and on-center control tests, suggesting that a general labyrinthine mechanism independent of gravity state is unlikely. Quantification of individual synaptic bodies in utricular hair cells in each group is a work in progress. Functional Considerations and General Conclusions: A change in gravity level has a profound effect on how an organism maintains equilibrium, and the neural response might involve the peripheral otolith receptors, the brain or both. In turn, the change of the sensitivity of peripheral otolith receptors in altered G may depend on the change of physical parameters of the otolith structures responsible for transferring the dynamics of head into the dynamics of the hair cell bundle deformations, as well as on the change of properties of local peripheral neural structures, first of all, synapses. The early hypersensitivity was unexpected, and the underlying mechanism is unknown. It is possible that the altered gravity initiates first an upregulation in gain in response to a challenge, any challenge. In this scenario we would expect to observe hypersensitivity in the early stages of μG exposure as well. Regrettably, those data are lacking. One of the hypotheses considered in the project was formulated as follows: ”afferent adaptation mechanisms (unquote, to altered gravity, quote) may involve either self-learning processes confined to the hair cell or responses of the hair cell-afferent system to feedback signal(s) that originate from a misbalance of information coming from otolith and extra-otolith contributor(s), or a combination of both”. The underlying basic assumptions of the feedback hypothesis were provided in the previous report. The experimental findings indicate that sensory plasticity ideed exists in the utricle following exposure to altered gravity conditions. However the mechanisms of neural adaptation in these experiments are still not clear. Theoretical work is in progress, and its progress is dependent on available data. Experimental analysis is still in progress. This analysis includes both the electrophysiological data and synaptic body counts at the transmission electron microscopic level.

Bibliography Type:	Description: (Last Updated: 09/17/2021)
Abstracts for Journals and Proceedings	Boyle RD, Popova Y, Varelas J, Intravaia A, Mofrad A, Kondrachuk A. "Neurovestibular adaptation and re-adaptation in the utricular otolith following extended periods of 3g exposure and re-adaptation to 1g." American Society for Gravitational and Space Biology 23rd annual meeting, Moffett Field, CA, October 25-28, 2007. Gravitational and Space Biology. 2007 Oct 21(1):34. , Oct-2007
Abstracts for Journals and Proceedings	Boyle RD, Popova Y, Varelas J, Mofrad A, Kondrachuk A. "Neurovestibular adaptation in the utricular otolith following hypergravity exposure." Society for Neurosciences, San Diego, CA, 3-7 November 2007. Abstracts, Society for Neurosciences, November 2007. , Nov-2007
Abstracts for Journals and Proceedings	Kondrachuk A, Boyle R, Sirenko S. "Mechanical Sensitivity of Semicircular Canals and the Effect of the Density Difference of the Cupula and Endolymph on Cupular Dynamics." 9th International Conference on Dynamical Systems - Theory and Applications, Lodz, Poland; 17-20 December 2007. Abstracts, 9th International Conference on Dynamical Systems - Theory and Applications, December 2007. , Dec-2007
Abstracts for Journals and Proceedings	Kondrachuk AV, Boyle R. "Intralabyrinthine pressure and function of cupula-endolymphatic system." Life in Space for Life on Earth: Joint 29th Annual ISGP Meeting, the 10th ESA Life Sciences Symposium, the 24th annual ASGSB Meeting, and the ELGRA Symposium, Angers, France, June 2008. Abstract book, Life in Space for Life on Earth: Joint 29th Annual ISGP Meeting, the 10th ESA Life Sciences Symposium, the 24th annual ASGSB Meeting, and the ELGRA Symposium, Angers, France, June 2008. , Jun-2008
Abstracts for Journals and Proceedings	Boyle R, Popova Y, Varelas J, Mofrad A, Kondrachuk A. "Neurovestibular adaptation in the utricular otolith following extended periods of 3G exposure and re-adaptation to 1G." Life in Space for Life on Earth: Joint 29th Annual ISGP Meeting, the 10th ESA Life Sciences Symposium, the 24th annual ASGSB Meeting, and the ELGRA Symposium, Angers, France, June 2008. Abstract book, Life in Space for Life on Earth: Joint 29th Annual ISGP Meeting, the 10th ESA Life Sciences Symposium, the 24th annual ASGSB Meeting, and the ELGRA Symposium, Angers, France, June 2008. , Jun-2008
Abstracts for Journals and Proceedings	Boyle RD, Popova Y, Varelas J, Mofrad A, Kondrachuk A. "Neurovestibular adaptation in the utricle following extended periods of hypergravity exposure and re-adaptation to 1G." Society for Neurosciences meeting, Washington, DC, 15-21 November. Abstract, Society for Neurosciences meeting, Washington, DC, 15-21 November. , Nov-2008
Articles in Other Journals or Periodicals	Kondrachuk AV, Boyle R. "Intralabyrinthine pressure and function of cupula-endolymphatic system." J Gravit Physiol., in press, December 2008. , Dec-2008
Articles in Other Journals or Periodicals	Boyle R, Popova Y, Varelas J, Mofrad A, Kondrachuk A. "Neurovestibular adaptation in the utricular otolith following extended periods of 3G exposure and re-adaptation to 1G." J Gravit Physiol, in press, December 2008. , Dec-2008
Articles in Peer-reviewed Journals	Kondrachuk AV, Sirenko SP, Boyle R. "The model of cupular dynamics in the case of the difference of the cupula and endolymph densities." J Gravit Physiol. 2007 Jul;14(1):P73-4. PMID: 18372706 , Jul-2007
Articles in Peer-reviewed Journals	Kondrachuk AV, Shipov AA, Astakhova TG, Boyle RD. "[Modern trends in mathematical modeling of the hemispheric channels functions]." Aviakosm Ekolog Med. 2009 Mar-Apr;43(2):5-12. Review. Russian. PubMed PMID: 19621795 , Mar-2009
Articles in Peer-reviewed Journals	Kondrachuk AV, Sirenko SP, Boyle R. "Effect of difference of cupula and endolymph densities on the dynamics of semicircular canal." J Vestib Res. 2008 Feb;18(2-3):69-88. PubMed PMID: 19126978 , Feb-2008
Articles in Peer-reviewed Journals	Chartrand T, McCollum G, Hanes DA, Boyle RD. "Symmetries of a generic utricular projection: neural connectivity and the distribution of utricular information." J Math Biol. 2016 Feb;72(3):727-53. Epub 2015 Jun 10. https://doi.org/10.1007/s00285-015-0900-5 ; PubMed PMID: 26059813 , Feb-2016

Articles in Peer-reviewed Journals	Boyle R, Ehsanian R, Mofrad A, Popova Y, Varelas J. "Morphology of the utricular otolith organ in the toadfish, Opsanus tau." J Comp Neurol. 2018 Jun 15;526(9):1571-88. Epub 2018 Mar 23. https://doi.org/10.1002/cne.24429 ; PubMed PMID: 29524209; PubMed Central PMCID: PMC5899691 , Jun-2018
Articles in Peer-reviewed Journals	Boyle R, Popova Y, Varelas J. "Influence of magnitude and duration of altered gravity and readaptation to 1 g on the structure and function of the utricle in toadfish, Opsanus tau." Front Physiol. 2018 Oct 22;9:1469. eCollection 2018. https://doi.org/10.3389/fphys.2018.01469 ; PubMed PMID: 30405430; PubMed Central PMCID: PMC6204554 , Oct-2018
Articles in Peer-reviewed Journals	Boyle R. "Otolith adaptive responses to altered gravity." Neurosci Biobehav Rev. 2021 Mar;122:218-28. https://doi.org/10.1016/j.neubiorev.2020.10.025 ; PMID: 33152424 , Mar-2021
Articles in Peer-reviewed Journals	Boyle R, Varelas J. "Otoconia structure after short- and long-duration exposure to altered gravity." J Assoc Res Otolaryngol. 2021 May 18. Ahead of print. https://doi.org/10.1007/s10162-021-00791-6 ; PMID: 34008038 , May-2021